

Pazorene 200mg Film Coated Tablets
Pazorene 400mg Film Coated Tablets

1. NAME OF THE MEDICINAL PRODUCT

Pazorene 200mg Film Coated Tablets
Pazorene 400mg Film Coated Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Pazorene 200mg Film Coated Tablets: Each tablet contains 216.7mg of Pazopanib hydrochloride equivalent to 200mg Pazopanib.

Pazorene 400mg Film Coated Tablets: Each tablet contains 433.4mg of Pazopanib hydrochloride equivalent to 400mg Pazopanib.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film-coated tablet.

Pazorene 200mg Film Coated Tablets: Grey, modified capsule-shaped, film-coated tablets, debossed with “P 200” on one side and plain on the other side.

Pazorene 400mg Film Coated Tablets: White, modified capsule-shaped, film-coated tablets, debossed with “P 400” on one side and plain on the other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

Renal cell carcinoma (RCC)

Pazorene is indicated for the treatment of patients with advanced and/or metastatic renal cell carcinoma (RCC).

Soft tissue sarcoma (STS)

Pazorene is indicated for the treatment of patients with advanced soft tissue sarcoma (STS) who have received prior chemotherapy.

The Phase III trial population excluded patients with gastrointestinal stromal tumor (GIST) or adipocytic STS.

4.2 Posology and Method of Administration

Dosage regimen

General target population

The recommended dose of Pazorene is 800 mg orally once daily.

Dose modifications

Dose modification, either an increase or decrease in dose, should be in 200 mg increments in a stepwise fashion based on individual tolerability in order to manage adverse reactions. The daily dose of Pazorene should not exceed 800 mg.

Special populations**Renal impairment**

Renal impairment is not expected to have a clinically relevant effect on Pazorene pharmacokinetics given the low renal excretion of pazopanib and metabolites. Renal impairment is not expected to influence pazopanib exposure, and dose adjustment is not necessary in patients with creatinine clearance ≥ 30 mL/min. There is no experience of pazopanib in patients with severe renal impairment or in patients undergoing peritoneal dialysis or hemodialysis; therefore, use of Pazorene is not recommended in these patients.

Hepatic impairment

The safety and pharmacokinetics of Pazorene in patients with pre-existing hepatic impairment have not been fully established.

No dose adjustment is required in patients with mild hepatic impairment as defined by alanine aminotransferase (ALT) and bilirubin.

The dose of Pazorene should be reduced to 200 mg per day in patients with moderate hepatic impairment.

There are insufficient data in patients with severe hepatic impairment (total bilirubin > 3 times the upper limit of normal [X ULN] regardless of the ALT value); therefore, use of Pazorene is not recommended in these patients.

Pediatric patients (below 18 years)

Pazorene is not recommended for use in children and adolescents under 18 years.

Geriatric patients (above 65 years)

No alteration of dosage, dosing frequency or route of administration is required in patients over 65 years.

Method of administration

Pazorene should be taken without food (at least one hour before or two hours after a meal). Pazorene should be taken whole with water and must not be broken or crushed. If a dose is missed, it should not be taken if it is less than 12 hours until the next dose.

4.3 Contraindications

Pazorene is contraindicated in patients with hypersensitivity to any of the ingredients.

4.4 Special Warnings and Precautions for Use**Hepatic effects**

Cases of hepatic failure (including fatalities) have been reported during use of pazopanib. In a

study with pazopanib, increase in serum transaminases (ALT, aspartate aminotransferase [AST]) and bilirubin were observed. In the majority of the cases, isolated increases in ALT and AST have been reported, without concomitant elevations of alkaline phosphatase or bilirubin. Patients over 60 years of age may be at greater risk for ALT >3 X ULN. Patients who carry the *HLA-B*57:01* allele also have an increased risk of pazopanib-associated ALT elevations. Liver function should be monitored in all subjects receiving pazopanib, regardless of genotype or age. The vast majority (over 90%) of all transaminase elevations of any grade occurred in the first 18 weeks. Grades are based on the National Cancer Institute Common Terminology Criteria for Adverse Events, Version 3 (NCI CTCAE).

Serum liver tests should be performed before initiation of treatment with pazopanib, at weeks 3, 5, 7 and 9, then at Months 3 and 4, with additional tests as clinically indicated. Periodic testing should then continue after Month 4.

The following guidelines are provided for patients with baseline values of total bilirubin ≤ 1.5 X ULN and AST and ALT ≤ 2 X ULN.

- Patients with isolated ALT elevations between 3 X ULN and 8 X ULN may be continued on pazopanib with weekly monitoring of liver function until ALT return to Grade 1 (NCI CTCAE) or baseline.
- Patients with ALT of > 8 X ULN should have pazopanib interrupted until they return to Grade 1 (NCI CTCAE) or baseline. If the potential benefit of reinitiating pazopanib treatment is considered to outweigh the risk for hepatotoxicity, then reintroduce pazopanib at a reduced dose of 400 mg once daily and perform serum liver tests weekly for 8 weeks. Following reintroduction of pazopanib, if ALT elevations > 3 X ULN recur, then pazopanib should be permanently discontinued.
- If ALT elevations > 3 X ULN occur concurrently with bilirubin elevations > 2 X ULN pazopanib should be permanently discontinued. Patients should be monitored until return to Grade 1 (NCI CTCAE) or baseline. pazopanib is a UGT1A1 inhibitor. Mild, indirect (unconjugated) hyperbilirubinemia may occur in patients with Gilbert's syndrome. Patients with only a mild indirect hyperbilirubinemia, known or suspected Gilbert's syndrome, and elevation in ALT > 3 X ULN should be managed as per the recommendations outlined for isolated ALT elevations.

Concomitant use of pazopanib and simvastatin increases the risk of ALT elevations and should be undertaken with caution and close monitoring.

Beyond recommending that patients with mild hepatic impairment are treated with 800 mg pazopanib once daily and reducing the initial starting dose to 200 mg per day for patients with moderate impairment, no further dose modification guidelines based on results of serum liver tests during therapy have been established for patients with pre-existing hepatic impairment.

Hypertension

In a study with pazopanib, events of hypertension including hypertensive crisis have occurred. Blood pressure should be well controlled prior to initiating pazopanib. Patients should be monitored for hypertension early after starting treatment (no longer than one week after starting pazopanib) and frequently thereafter to ensure blood pressure control, and treated promptly with a combination of standard anti-hypertensive therapy and pazopanib dose reduction or interruption as clinically warranted. Hypertension (systolic blood pressure ≥ 150 mm Hg or diastolic blood pressure ≥ 100 mm Hg) occurs early in the course of

pazopanib treatment (approximately 40% of cases occurred by Day 9 and approximately 90% of cases occurred in the first 18 weeks). Pazopanib should be discontinued if there is evidence of hypertensive crisis or if hypertension is severe and persists despite anti-hypertensive therapy and pazopanib dose reduction.

Posterior reversible encephalopathy syndrome (PRES)/Reversible posterior leukoencephalopathy syndrome (RPLS)

PRES/RPLS has been reported in association with pazopanib. PRES/RPLS can present with headache, hypertension, seizure, lethargy, confusion, blindness and other visual and neurological disturbances, and can be fatal. Pazopanib should be permanently discontinued in patients developing PRES/RPLS.

Interstitial lung disease (ILD)/Pneumonitis

ILD, which can be fatal, has been reported in association with pazopanib. Patients should be monitored for pulmonary symptoms indicative of ILD/pneumonitis and pazopanib should be discontinued in patients developing ILD or pneumonitis.

Cardiac dysfunction

In a study with Pazopanib, events of cardiac dysfunction such as congestive heart failure and decreased left ventricular ejection fraction (LVEF) have occurred. In a randomized RCC trial of Pazorene compared with sunitinib, in subjects who had baseline and follow-up LVEF measurements, myocardial dysfunction was observed in 13% (47/362) of subjects in the Pazorene arm compared to 11% (42/369) of subjects in the sunitinib arm. Congestive heart failure was observed in 0.5% of subjects in each treatment arm. In the Phase III STS clinical trial, congestive heart failure was reported in 3 out of 240 subjects (1%). In this trial decreases in LVEF in subjects who had post-baseline measurement were detected in 11% (16/142) in the Pazorene arm compared with 5% (2/40) in the placebo arm. Fourteen of the 16 subjects in the Pazorene arm had concurrent hypertension which may have exacerbated cardiac dysfunction in patients at risk (e.g., those with prior anthracycline therapy) by increasing cardiac after-load. Blood pressure should be monitored and managed promptly using a combination of anti-hypertensive therapy and dose modification of Pazopanib (interruption and re-initiation at a reduced dose based on clinical judgment). Patients should be carefully monitored for clinical signs or symptoms of congestive heart failure. Baseline and periodic evaluation of LVEF is recommended in patients at risk of cardiac dysfunction.

QT prolongation and torsade de pointes

In a study with pazopanib, events of QT prolongation or torsade de pointes have occurred. Pazopanib should be used with caution in patients with a history of QT interval prolongation, in patients taking antiarrhythmics or other medications that may potentially prolong QT interval, or in patients with relevant pre-existing cardiac disease. When using pazopanib, baseline and periodic monitoring of electrocardiograms and maintenance of electrolytes (calcium, magnesium, potassium) within normal range is recommended.

Arterial thrombotic events

In a study with pazopanib, myocardial infarctions, angina, ischemic stroke and transient ischemic attack were observed. Fatal events have been observed. Pazopanib should be used with caution in patients who are at increased risk of thrombotic events or who have had a history of thrombotic events. Pazopanib has not been studied in patients who have had an event within the previous 6 months. A treatment decision should be made based on the assessment of individual patient's benefit/risk.

Venous thromboembolic events

In a study with pazopanib, venous thromboembolic events including venous thrombosis and fatal pulmonary embolus have occurred. The incidence was higher in the STS population than in the RCC population.

Thrombotic microangiopathy (TMA)

Thrombotic microangiopathy (TMA) has been reported in a study of pazopanib as monotherapy, in combination with bevacizumab, and in combination with topotecan. Pazopanib should be permanently discontinued in patients developing TMA. Reversal of effects of TMA has been observed after treatment was discontinued. Pazopanib is not indicated for use in combination with other agents.

Hemorrhagic events

In a study with pazopanib, hemorrhagic events have been reported. Fatal hemorrhagic events have occurred. Pazopanib has not been studied in patients who had a history of hemoptysis, cerebral hemorrhage, or clinically significant gastrointestinal hemorrhage in the past 6 months. Pazopanib should be used with caution in patients with significant risk of hemorrhage.

Aneurysms and artery dissections

Artery dissections and aneurysms have been reported in association with VEGF pathway inhibitors, including pazopanib. The use of VEGF pathway inhibitors in patients with or without hypertension may promote the formation of aneurysm and/or artery dissections. Before initiating pazopanib, this risk should be carefully considered in patients with risk factors such as hypertension or history of aneurysm.

Gastrointestinal perforations and fistula

In a study with pazopanib, events of gastrointestinal (GI) perforation or fistula have occurred. Fatal perforation events have occurred. Pazopanib should be used with caution in patients at risk for GI perforation or fistula.

Wound healing

No formal studies of the effect of pazopanib on wound healing have been conducted. Since vascular endothelial growth factor (VEGF) inhibitors may impair wound healing, treatment with pazopanib should be stopped at least 7 days prior to scheduled surgery. The decision to resume pazopanib after surgery should be based on clinical judgement of adequate wound healing. Pazopanib should be discontinued in patients with wound dehiscence.

Hypothyroidism

In a study with pazopanib, events of hypothyroidism have occurred. Proactive monitoring of thyroid function tests is recommended.

Proteinuria

In a study with pazopanib, proteinuria has been reported. Baseline and periodic urinalyses during treatment are recommended and patients should be monitored for worsening proteinuria. Pazopanib should be discontinued if the patient develops nephrotic syndrome.

Tumor lysis syndrome (TLS)

Cases of TLS, including fatal cases, have been reported in patients treated with pazopanib.

Patients generally at risk of TLS are those with rapidly growing tumors, a high tumor burden, renal dysfunction, or dehydration. Preventative measures such as treatment of high uric acid levels and intravenous hydration should be considered prior to initiation of pazopanib. Patients at risk should be closely monitored and treated as clinically indicated.

Infections

Cases of serious infections (with or without neutropenia), in some cases with fatal outcome, have been reported.

Combination with other systemic anti-cancer therapies

A study of pazopanib in combination with pemetrexed (non-small cell lung cancer (NSCLC)), lapatinib (cervical cancer) or pembrolizumab (advanced renal cell carcinoma) were terminated early due to concerns over increased toxicity and/or mortality, and a safe and effective combination dose has not been established with these regimens. Pazopanib is not indicated for use in combination with other anti-cancer agents.

Juvenile animal toxicity

Because the mechanism of action of pazopanib can severely affect organ growth and maturation during early post-natal development, pazopanib should not be given to human pediatric patients younger than 2 years of age.

Pregnancy

Studies in animals have shown reproductive toxicity.

Based on animal reproduction studies and its mechanism of action, pazopanib can cause fetal harm when administered to a pregnant woman. Pregnant women should be advised of the potential risk to a fetus. Females of reproductive potential should be advised to avoid becoming pregnant while receiving treatment with pazopanib.

Interactions

Concomitant treatment with strong inhibitors of CYP3A4, P-glycoprotein (P-gp) or breast cancer resistance protein (BCRP) should be avoided due to risk of increased exposure to pazopanib. Selection of alternative concomitant medicinal products with no or minimal potential to inhibit CYP3A4, P-gp or BCRP should be considered.

4.5 Interaction with Other Medicinal Products and Other Forms of Interaction

Drugs that inhibit or induce cytochrome P450 3A4 enzymes

In studies suggested that the oxidative metabolism of pazopanib in human liver microsomes is mediated primarily by CYP3A4, with minor contributions from CYP1A2 and CYP2C8. Therefore, inhibitors and inducers of CYP3A4 may alter the metabolism of pazopanib.

CYP3A4, P-gp, BCRP inhibitors

Pazopanib is a substrate for CYP3A4, P-gp and BCRP.

Concurrent administration of Pazopanib (400 mg once daily) with the strong CYP3A4 and P-gp inhibitor, ketoconazole (400 mg once daily) for 5 consecutive days, resulted in a 66% and 45% increase in mean Pazopanib AUC₍₀₋₂₄₎ and C_{max}, respectively, relative to administration of pazopanib alone (400 mg once daily for 7 days). Pazopanib C_{max} and AUC increase in a less than dose proportional fashion with increasing dose over the range of 50 mg to 2000 mg.

Therefore, a dose reduction to 400 mg pazopanib once daily in the presence of strong CYP3A4 inhibitors will, in the majority of patients, result in systemic exposure similar to that observed after administration of 800 mg pazopanib once daily alone. Some patients however may have systemic pazopanib exposure greater than what has been observed after administration of 800 mg pazopanib alone.

Co-administration of pazopanib with other strong inhibitors of the CYP3A4 family (e.g., itraconazole, clarithromycin, atazanavir, indinavir, nefazodone, nelfinavir, ritonavir, saquinavir, telithromycin, voriconazole) may increase pazopanib concentrations. Grapefruit juice may also increase plasma concentrations of pazopanib.

Administration of 1500 mg lapatinib a substrate and weak inhibitor of CYP3A4, P-gp and BCRP with 800 mg pazopanib resulted in an approximately 50% to 60% increase in mean pazopanib AUC₍₀₋₂₄₎ and C_{max} compared to administration of 800 mg pazopanib alone. Co-administration of pazopanib with a CYP3A4, P-gp, and BCRP inhibitor, such as lapatinib, will result in an increase in plasma pazopanib concentrations.

Concomitant use of pazopanib with a strong CYP3A4 inhibitor should be avoided. If no medically acceptable alternative to a strong CYP3A4 inhibitor is available, the dose of pazopanib should be reduced to 400 mg daily during concomitant administration. Further dose reduction may be considered if possible drug-related adverse events are observed.

Combination with strong P-gp or BCRP inhibitors should be avoided, or selection of an alternate concomitant medication with no or minimal potential to inhibit P-gp or BCRP is recommended.

CYP3A4 inducers

CYP3A4 inducers such as rifampin may decrease plasma pazopanib concentrations. Selection of an alternative concomitant medication with no or minimal enzyme induction potential is recommended.

Effects of Pazopanib on CYP substrates

In studies with human liver microsomes showed that pazopanib inhibited CYP enzymes 1A2, 3A4, 2B6, 2C8, 2C9, 2C19, and 2E1. Potential induction of human CYP3A4 was demonstrated in an in vitro human PXR assay. Studies, using pazopanib 800 mg once daily, have demonstrated that pazopanib does not have a clinically relevant effect on the pharmacokinetics of caffeine (CYP1A2 probe substrate), warfarin (CYP2C9 probe substrate), or omeprazole (CYP2C19 probe substrate) in cancer patients. Pazopanib resulted in an increase of approximately 30% in the mean AUC and C_{max} of midazolam (CYP3A4 probe substrate) and increases of 33% to 64% in the ratio of dextromethorphan to dextrothorphan concentrations in the urine after oral administration of dextromethorphan (CYP2D6 probe substrate). Co-administration of pazopanib 800 mg once daily and paclitaxel 80 mg/m² (CYP3A4 and CYP2C8 substrate) once weekly resulted in a mean increase of 26% and 31% in paclitaxel AUC and C_{max}, respectively.

Effects of Pazopanib on other enzymes and transporters

In studies also showed that pazopanib is a potent inhibitor of UGT1A1 and OATP1B1 with IC₅₀ of 1.2 and 0.79 microM, respectively. Pazopanib may increase concentrations of drugs primarily eliminated through UGT1A1 and OATP1B1.

Effect of concomitant use of Pazopanib and simvastatin

Concomitant use of pazopanib and simvastatin increases the incidence of ALT elevations. Across monotherapy studies with pazopanib, ALT >3X ULN was reported in 126/895 (14%) of patients who did not use statins, compared with 11/41 (27%) of patients who had concomitant use of simvastatin ($p=0.038$). If a patient receiving concomitant simvastatin develops ALT elevations, follow guidelines for pazopanib posology and discontinue simvastatin. Insufficient data are available to assess the risk of concomitant administration of alternative statins and pazopanib.

Drug-food/drink interactions

Administration of pazopanib with a high-fat or low-fat meal results in an approximately 2-fold increase in AUC and C_{max} . Therefore, pazopanib should be administered at least 1 hour before or 2 hours after a meal.

Medicines that raise gastric pH

Concomitant administration of pazopanib with esomeprazole decreases the bioavailability of pazopanib by approximately 40% (AUC and C_{max}), and co-administration of pazopanib with medicines that increase gastric pH should be avoided.

4.6 Fertility, pregnancy and lactation**Pregnancy**

Based on animal reproduction studies and its mechanism of action, Pazopanib can cause fetal harm when administered to a pregnant woman. There are no adequate data from the use of Pazopanib in pregnant women. In animal developmental toxicity studies, oral administration of pazopanib to pregnant rats and rabbits throughout organogenesis resulted in teratogenicity and abortion at systemic exposures lower than that observed at the maximum recommended human dose of 800 mg/day (based on AUC). Pazopanib should not be used during pregnancy unless the clinical condition of the woman requires treatment with pazopanib. Pregnant women or females of reproductive potential should be advised of the potential risk to a fetus.

Animal data

In a female fertility and early embryonic development study in rats, post-implantation loss, embryo lethality and decreased fetal body weights were noted at dosages ≥ 10 mg/kg/day (approximately 0.2-fold the AUC at the MRHD of 800 mg/day) and increased pre-implantation loss and early resorptions were noted at dosages ≥ 30 mg/kg/day (approximately 0.4-fold the AUC at the MRHD of 800 mg/day).

In embryo-fetal development toxicity studies, pazopanib produced teratogenic effects (including cardiovascular malformations), delayed ossification, increased post-implantation loss, reduced fetal body weight and embryo lethality in rats at a dose level of ≥ 3 mg/kg/day (approximately 0.1-fold the AUC at the MRHD of 800 mg/day). In rabbits, maternal toxicity (body weight loss, reduced food consumption), increased post-implantation loss and abortion were observed at doses ≥ 30 mg/kg/day (approximately 0.007-fold the AUC at the MRHD of 800 mg/day), while fetal weight was reduced at doses ≥ 3 mg/kg/day (AUC not calculated).

Lactation

There is no information regarding the presence of pazopanib or its metabolites in human milk, or their effects on the breastfed infant, or on milk production. Because of the potential

for serious adverse reactions in breastfed infants from pazopanib, a lactating woman should be advised not to breastfeed during treatment with pazopanib.

Fertility

Contraception

Females

Females of reproductive potential should be advised to use effective contraception during treatment with pazopanib and for at least 2 weeks after the last dose.

Males

Male patients (including those who have had vasectomies) with female partners who are pregnant, possibly pregnant, or who could become pregnant should use condoms while taking pazopanib and for at least 2 weeks after the last dose.

Infertility

Based on findings from animal studies, pazopanib may impair fertility in males and females of reproductive potential while receiving treatment.

4.7 Effects on Ability to Drive and Use Machines

Pazorene has no or negligible influence on the ability to drive and use machines. A detrimental effect on such activities cannot be predicted from the pharmacology of pazopanib. The clinical status of the patient and the adverse event profile of pazopanib should be borne in mind when considering the patient's ability to perform tasks that require judgement, motor or cognitive skills. Patients should avoid driving or using machines if they feel dizzy, tired or weak.

4.8 Undesirable Effects

Summary of the safety profile

The most important serious adverse reactions identified in the RCC or STS studies were transient ischaemic attack, ischaemic stroke, myocardial ischaemia, myocardial and cerebral infarction, cardiac dysfunction, gastrointestinal perforation and fistula, QT prolongation, Torsade de Pointes and pulmonary, gastrointestinal and cerebral haemorrhage. Other important serious adverse reactions identified in STS studies included venous thromboembolic events, left ventricular dysfunction and pneumothorax.

Fatal events that were considered possibly related to pazopanib included gastrointestinal haemorrhage, pulmonary haemorrhage/haemoptysis, abnormal hepatic function, intestinal perforation and ischaemic stroke.

The most common adverse reactions in RCC and STS studies included: diarrhoea, hair colour change, skin hypopigmentation, exfoliative rash, hypertension, nausea, headache, fatigue, anorexia, vomiting, dysgeusia, stomatitis, weight decreased, pain, elevated alanine aminotransferase and elevated aspartate aminotransferase.

Adverse drug reactions, all grades, which were reported in RCC and STS studies or during the post-marketing period are listed below by MedDRA body system organ class, frequency

and grade of severity. The following convention has been utilised for the classification of frequency: very common; common; uncommon; rare; very rare; and not known.

Tabulated list of adverse reactions

Table 1: Treatment-related adverse reactions reported in RCC studies or during post-marketing period

System Organ Class	Frequency	Adverse reactions
Infections and Infestations	Common	Infections (with or without neutropenia) [†]
	Uncommon	Gingival infection
		Infectious peritonitis
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	Uncommon	Tumour pain
Blood and lymphatic system disorders	Common	Thrombocytopenia
		Neutropenia
		Leukopenia
	Uncommon	Polycythaemia
	Rare	Thrombotic microangiopathy (including thrombotic thrombocytopenic purpura and haemolytic uraemic syndrome) [†]
Endocrine disorders	Common	Hypothyroidism
Metabolism and nutrition disorders	Very common	Decreased appetite ^e
	Common	Hypophosphataemia
		Dehydration
	Uncommon	Hypomagnesaemia
	Not known	Tumour lysis syndrome*
Psychiatric disorders	Common	Insomnia
Nervous system disorders	Very common	Dysgeusia ^c
		Headache
	Common	Dizziness
		Lethargy
		Paraesthesia
		Peripheral sensory neuropathy
		Hypoaesthesia
	Uncommon	Transient ischaemic attack
		Somnolence
		Cerebrovascular accident
		Ischaemic stroke
	Rare	Posterior reversible encephalopathy / reversible posterior leukoencephalopathy syndrome [†]
Eye disorders	Common	Vision blurred
	Uncommon	Retinal detachment [†]
		Retinal tear [†]
		Eyelash discolouration
Cardiac disorders	Uncommon	Bradycardia

		Myocardial infarction
		Cardiac dysfunction ^f
		Myocardial ischaemia
Vascular disorders	Very common	Hypertension
	Common	Hot flush
		Venous thromboembolic event ^g
		Flushing
	Uncommon	Hypertensive crisis
		Haemorrhage
	Rare	Aneurysms and artery dissections [†]
Respiratory, thoracic and mediastinal disorders	Common	Epistaxis
		Dysphonia
		Dyspnoea
		Haemoptysis
	Uncommon	Rhinorrhoea
		Pulmonary haemorrhage
		Pneumothorax
	Rare	Interstitial lung disease/pneumonitis [†]
Gastrointestinal disorders	Very common	Diarrhoea
		Nausea
		Vomiting
		Abdominal pain ^a
	Common	Stomatitis
		Dyspepsia
		Flatulence
		Abdominal distension
		Mouth ulceration
		Dry mouth
	Uncommon	Pancreatitis
		Rectal haemorrhage
		Haematochezia
		Gastrointestinal haemorrhage
		Melaena
		Frequent bowel movements
		Anal haemorrhage
		Large intestine perforation
		Mouth haemorrhage
		Upper gastrointestinal haemorrhage
		Enterocutaneous fistula
		Haematemesis
		Haemorrhoidal haemorrhage
		Ileal perforation
		Oesophageal haemorrhage
		Retroperitoneal haemorrhage
Hepatobiliary disorders	Common	Hyperbilirubinaemia
		Hepatic function abnormal
		Hepatotoxicity
	Uncommon	Jaundice

		Drug induced liver injury
		Hepatic failure†
Skin and subcutaneous disorders	Very common	Hair colour change
		Palmar-plantar erythrodysaesthesia syndrome
		Alopecia
		Rash
	Common	Skin hypopigmentation
		Dry skin
		Pruritus
		Erythema
		Skin depigmentation
		Hyperhidrosis
	Uncommon	Nail disorders
		Skin exfoliation
		Photosensitivity reaction
		Rash erythematous
		Skin disorder
		Rash macular
		Rash pruritic
		Rash vesicular
		Pruritus generalised
		Rash generalised
		Rash papular
		Plantar erythema
		Skin ulcer†
Musculoskeletal and connective tissue disorders	Common	Arthralgia
		Myalgia
		Muscle spasms
	Uncommon	Musculoskeletal pain
Renal and urinary disorders	Very Common	Proteinuria
	Uncommon	Haemorrhage urinary tract
Reproductive system and breast disorders	Uncommon	Menorrhagia
		Vaginal haemorrhage
		Metrorrhagia
General disorders and administration site conditions	Very common	Fatigue
	Common	Mucosal inflammation
		Asthenia
		Oedema ^b
		Chest pain
	Uncommon	Chills
		Mucous membrane disorder
Investigations	Very common	Alanine aminotransferase increased
		Aspartate aminotransferase increased
	Common	Weight decreased
		Blood bilirubin increased
		Blood creatinine increased
		Lipase increased
		White blood cell count decreased ^d

		Blood thyroid stimulating hormone increased
		Amylase increased
		Gamma-glutamyltransferase increased
		Blood pressure increased
		Blood urea increased
		Liver function test abnormal
	Uncommon	Hepatic enzyme increased
		Blood glucose decreased
		Electrocardiogram QT prolonged
		Transaminase increased
		Thyroid function test abnormal
		Blood pressure diastolic increased
		Blood pressure systolic increased

† Treatment-related adverse reaction reported during post-marketing period (spontaneous case reports and serious adverse reactions from all pazopanib clinical studies).

* Treatment-related adverse reaction reported only during the post-marketing period. Frequency cannot be estimated from the available data.

The following terms have been combined:

- a Abdominal pain, abdominal pain upper and abdominal pain lower
- b Oedema, oedema peripheral, eye oedema, localised oedema and face oedema
- c Dysgeusia, ageusia and hypogeusia
- d White cell count decreased, neutrophil count decreased and leukocyte count decreased
- e Decreased appetite and anorexia
- f Cardiac dysfunction, left ventricular dysfunction, cardiac failure and restrictive cardiomyopathy
- g Venous thromboembolic event, deep vein thrombosis, pulmonary embolism and thrombosis

Neutropenia, thrombocytopenia and palmar-plantar erythrodysaesthesia syndrome were observed more frequently in patients of East Asian descent.

Table 2 Treatment-related adverse reactions reported in STS studies or during post-marketing period

System Organ Class	Frequency	Adverse reactions
Infections and infestations	Common	Gingival infection
Neoplasms benign, malignant and unspecified (incl cysts and polyps)	Very common	Tumour pain
Blood and lymphatic system disorders^f	Very common	Leukopenia
		Thrombocytopenia
		Neutropenia
	Uncommon	Thrombotic microangiopathy (including thrombotic thrombocytopenic purpura and haemolytic uraemic syndrome)
Endocrine disorders	Common	Hypothyroidism
Metabolism and nutrition disorders	Very common	Decreased appetite
		Hypoalbuminemia ^f

	Common	Dehydration
	Uncommon	Hypomagnesaemia
	Not known	Tumour lysis syndrome*
Psychiatric disorders	Common	Insomnia
Nervous system disorders	Very common	Dysgeusia ^c
		Headache
	Common	Peripheral sensory neuropathy
		Dizziness
	Uncommon	Somnolence
		Paresthesia
Cerebral infarction		
Eye disorders	Common	Vision blurred
Cardiac disorders		Cardiac dysfunction ^g
		Left ventricular dysfunction
		Bradycardia
	Uncommon	Myocardial infarction
Vascular disorders	Very common	Hypertension
	Common	Venous thromboembolic event ^d
		Hot flush
		Flushing
	Uncommon	Haemorrhage
Rare	Aneurysms and artery dissections	
Respiratory, thoracic and mediastinal disorders	Common	Epistaxis
		Dysphonia
		Dyspnoea
		Cough
		Pneumothorax
		Hiccups
		Pulmonary haemorrhage
	Uncommon	Oropharyngeal pain
		Bronchial haemorrhage
		Rhinorrhoea
		Haemoptysis
Rare	Interstitial lung disease/pneumonitis [†]	
Gastrointestinal disorders	Very common	Diarrhoea
		Nausea
		Vomiting
		Abdominal pain ^a
		Stomatitis
	Common	Abdominal distension
		Dry mouth
		Dyspepsia
		Mouth haemorrhage
		Flatulence
		Anal haemorrhage
	Uncommon	Gastrointestinal haemorrhage
Rectal haemorrhage		

		Enterocutaneous fistula
		Gastric haemorrhage
		Melaena
		Oesophageal haemorrhage
		Peritonitis
		Retroperitoneal haemorrhage
		Upper gastrointestinal haemorrhage
		Ileal perforation
Hepatobiliary disorders	Uncommon	Hepatic function abnormal
	Not known	Hepatic failure*
Skin and subcutaneous disorders	Very common	Hair colour change
		Skin hypopigmentation
		Exfoliative rash
	Common	Alopecia
		Skin disorder ^c
		Dry skin
		Hyperhydrosis
		Nail disorder
		Pruritus
		Erythema
	Uncommon	Skin ulcer
		Rash
		Rash papular
		Photosensitivity reaction
		Palmar-plantar erythrodysesthesia syndrome
Musculoskeletal and connective tissue disorders	Common	Musculoskeletal pain
		Myalgia
		Muscle spasms
	Uncommon	Arthralgia
Renal and urinary disorders	Uncommon	Proteinuria
Reproductive system and breast disorder	Uncommon	Vaginal haemorrhage
		Menorrhagia
General disorders and administration site conditions	Very common	Fatigue
	Common	Oedema ^b
		Chest pain
		Chills
	Uncommon	Mucosal inflammation ^c
		Asthenia
Investigations^h	Very common	Weight decreased
	Common	Ear, nose and throat examination abnormal ^c
		Alanine aminotransferase increased
		Blood cholesterol abnormal
		Aspartate aminotransferase increased
		Gamma glutamyltransferase increased
	Uncommon	Blood bilirubin increased
		Aspartate aminotransferase
		Alanine aminotransferase

		Platelet count decreased
		Electrocardiogram QT prolonged

† Treatment-related adverse reaction reported during post-marketing period (spontaneous case reports and serious adverse reactions from all pazopanib clinical studies).

* Treatment-related adverse reaction reported only during the post-marketing period. Frequency cannot be estimated from the available data.

The following terms have been combined:

- a Abdominal pain, abdominal pain upper and gastrointestinal pain
- b Oedema, oedema peripheral and eyelid oedema
- c The majority of these cases were Palmar-plantar erythrodysaesthesia syndrome
- d Venous thromboembolic events – includes Deep vein thrombosis, Pulmonary embolism and Thrombosis terms
- e The majority of these cases describe mucositis
- g Cardiac dysfunction events – includes Left ventricular dysfunction, Cardiac failure and Restrictive cardiomyopathy
- h Frequency is based on adverse events reported by investigators. Laboratory abnormalities were reported as adverse events less frequently by investigators than as indicated by laboratory value tables.

Neutropenia, thrombocytopenia and palmar-plantar erythrodysaesthesia syndrome were observed more frequently in patients of East Asian descent.

Paediatric population

The safety profile in paediatric patients was similar to that reported with pazopanib in adults in the approved indications

4.9 Overdose

Pazopanib doses up to 2,000 mg daily have been evaluated in studies. Grade 3 fatigue (dose limiting toxicity) and Grade 3 hypertension were each observed in 1 of 3 patients dosed at 2,000 mg and 1,000 mg daily, respectively.

Symptoms and signs

There is currently limited experience with overdosage in pazopanib.

Treatment

Further management should be as clinically indicated or as recommended by the national poisons center, where available. Hemodialysis is not expected to enhance the elimination of pazopanib because pazopanib is not significantly renally excreted and is highly bound to plasma proteins.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Pharmacotherapeutic group, ATC

Pharmacotherapeutic group: Antineoplastic agents – Protein kinase inhibitor, ATC Code: L01XE11.

Mechanism of action (MOA)

Pazopanib is an orally administered, potent multi-target tyrosine kinase inhibitor (TKI) of vascular endothelial growth factor receptors (VEGFR)-1, -2, and -3, platelet-derived growth factor (PDGFR)-alpha and -beta, and stem cell factor receptor (c-KIT), with IC₅₀ values of 10, 30, 47, 71, 84 and 74 nM, respectively. In preclinical experiments, pazopanib dose-dependently inhibited ligand-induced auto-phosphorylation of VEGFR-2, c-Kit and PDGFR-beta receptors in cells. *In vivo*, pazopanib inhibited VEGF-induced VEGFR-2 phosphorylation in mouse lungs, angiogenesis in various animal models, and the growth of multiple human tumor xenografts in mice.

5.2 Pharmacokinetic Properties

Absorption

Pazopanib is absorbed orally with median time to achieve peak concentrations of 2.0 to 4.0 hours after the dose. Daily dosing results in 1.23- to 4-fold increase in AUC. There was no consistent increase in AUC and C_{max} when the pazopanib dose increased above 800 mg once daily.

Systemic exposure to pazopanib is increased when administered with food. Administration of pazopanib with a high-fat or low-fat meal results in an approximately 2-fold increase in AUC and C_{max}. Therefore, pazopanib should be administered at least 1 hour before or 2 hours after a meal.

Administration of a single pazopanib 400 mg crushed tablet increased AUC₍₀₋₇₂₎ by 46% and C_{max} by approximately 2-fold and decreased t_{max} by approximately 1.5 hours compared to administration of the whole tablet. These results indicate that the bioavailability and the rate of pazopanib oral absorption are increased after administration of the crushed tablet relative to administration of the whole tablet. Therefore, due to this potential for increased exposure, tablets should not be crushed.

Distribution

Binding of pazopanib to human plasma protein *in vivo* was greater than 99% with no concentration dependence over the range of 10 to 100 microgram/mL. In studies suggest that Pazopanib is a substrate for P-glycoprotein (P-gp) and breast cancer resistant protein (BCRP).

Biotransformation/metabolism

Results from studies demonstrated that the metabolism of pazopanib is mediated primarily by CYP3A4, with minor contributions from CYP1A2 and CYP2C8.

Elimination

Pazopanib is eliminated slowly with mean half-life of 30.9 hours after administration of the recommended dose of 800 mg. Elimination is primarily via feces with renal elimination accounting for <4% of the administered dose.

Special populations

Renal impairment

In a population pharmacokinetic analysis using 408 subjects with various cancers, creatinine clearance (30 to 150 mL/min) did not influence clearance of pazopanib. Renal impairment is not expected to influence pazopanib exposure, and dose adjustment is not necessary in patients

with creatinine clearance ≥ 30 mL/min.

Hepatic impairment

The median steady-state pazopanib $C_{\max}$ and $AUC_{(0-24)}$ in patients with mild hepatic impairment (defined as either normal bilirubin and any degree of ALT elevations or as an elevation of bilirubin up to 1.5 X ULN regardless of the ALT value) after a once daily dose of 800 mg/day (30.9 microgram/mL, range 12.5 to 47.3 and 841.8 microgram.hr/mL, range 600.4 to 1,078) are similar to the median in patients with no hepatic impairment (49.4 microgram/mL, range 17.1 to 85.7 and 888.2 microgram.hr/mL, range 345.5 to 1,482).

The maximally tolerated pazopanib dose (MTD) in patients with moderate hepatic impairment (defined as an elevation of bilirubin > 1.5 X to 3 X ULN regardless of the ALT values) was 200 mg once daily. The median steady-state values of $C_{\max}$ (22.4 microgram/mL, range 6.4 to 32.9) and $AUC_{(0-24)}$ (350.0 microgram.hr/mL, range 131.8 to 487.7) after administration of 200 mg pazopanib once daily in subjects with moderate hepatic impairment were approximately 45% and 39%, respectively, that of the corresponding median values after administration of 800 mg once daily in subjects with normal hepatic function.

There are insufficient data in patients with severe hepatic impairment (total bilirubin > 3 X ULN regardless of the ALT value); therefore, use of pazopanib is not recommended in these patients.

Pharmacogenomics

In a pharmacogenetic meta-analysis of data from 31 clinical studies of pazopanib administered either as monotherapy or in combination with other agents, ALT > 5 X ULN (NCI CTC Grade 3) occurred in 19% of HLA-B*57:01 allele carriers and in 10% of non-carriers. In this dataset, 133/2235 (6%) of the patients carried the HLA-B*57:01 allele.

6 PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Tablet core:

Microcrystalline cellulose, Povidone, Sodium Starch Glycolate, Magnesium stearate

Film coat:

Opadry Complete Film Coating System 15B675001 Grey (200 mg), Opadry Complete Film Coating System 15B680006 White (400 mg)

6.2 Incompatibilities

Not applicable

6.3 Shelf Life

Please refer to the expiry date on the product labels.

6.4 Special Precautions for Storage

Store below 30°C.

6.5 Nature and Content of Container

Pazorene 200mg Film Coated Tablets: High-density polyethylene (HDPE) bottles with child resistant polypropylene closures containing 30 or 90 tablets. Not all pack sizes will be marketed.

Pazorene 400mg Film Coated Tablets: High-density polyethylene (HDPE) bottles with child resistant polypropylene closures containing 30 or 60 tablets. Not all pack sizes will be marketed.

6.6 Special precautions for disposal and other handling

No special requirements.

7 MANUFACTURER

Manufactured by:

Novugen Oncology Sdn. Bhd.
No. 47, Jalan Lengkok Teknologi 2,
Taman Teknologi Enstek Fasa 1,
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Product Registration Holder:

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8 DATE OF REVISION

02/05/2024

Package insert is drafted in Times New Roman with font size of 11, subject to change based on any other legible font and font size.
